

SHETH L.U.J. & SIR M.V. COLLEGE OF SCIENCE
SUBJECT - Data Analysis with R/SAS/SPSS

Aim - Performing independent two-sample t-tests using t.test() with grouping (R).

Output :

```
> data <- read.csv("penguins.csv")
> head(data)
  species      island culmen_length culmen_depth flipper_length body_mass   sex
1 Adelie Torgersen    39.1         18.7         181        3750    MALE
2 Adelie Torgersen    39.5         17.4         186        3800    FEMALE
3 Adelie Torgersen    40.3         18.0         195        3250    FEMALE
4 Adelie Torgersen    NA          NA          NA          NA    <NA>
5 Adelie Torgersen    36.7         19.3         193        3450    FEMALE
6 Adelie Torgersen    39.3         20.6         190        3650    MALE
> str(data)
'data.frame':   344 obs. of  7 variables: "species" "island" "culmen_length" "culmen_depth" "flipper_length" "body_mass" "sex"
 $ species: chr  "Adelie" "Adelie" "Adelie" "Adelie" "Adelie" "Adelie" ...
 $ island : chr  "Torgersen" "Torgersen" "Torgersen" "Torgersen" "Torgersen" "Torgersen" ...
 $ culmen_length: num  39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42 ...
 $ culmen_depth : num  18.7 17.4 18 NA 19.3 20.6 17.8 19.6 18.1 20.2 ...
 $ flipper_length: int   181 186 195 NA 193 190 181 195 193 190 ...
 $ body_mass    : int   3750 3800 3250 NA 3450 3650 3625 4675 3475 4250 ...
 $ sex          : chr  "MALE" "FEMALE" "FEMALE" "MALE" "MALE" "MALE" ...
> data$group_culmen_length <- ifelse(data$culmen_length >= median(data$culmen_length, na.rm = TRUE), "High", "Low")
> data$group_culmen_depth <- ifelse(data$culmen_depth >= median(data$culmen_depth, na.rm = TRUE), "High", "Low")
> data$group_flipper_length <- ifelse(data$flipper_length >= median(data$flipper_length, na.rm = TRUE), "High", "Low")
> data$group_body_mass <- ifelse(data$body_mass >= median(data$body_mass, na.rm = TRUE), "High", "Low")
> table(data$group_culmen_length)

High Low
171 171
> table(data$group_culmen_depth)

High Low
176 166
> table(data$group_flipper_length)

High Low
176 166
> table(data$group_body_mass)

High Low
176 166
```

```
Welch Two Sample t-test

data: culmen_length by group_culmen_length
t = 31.413, df = 339.97, p-value < 2.2e-16
alternative hypothesis: true difference in means between group High and group Low is not equal to 0
95 percent confidence interval:
 8.814141 9.991707
sample estimates:
mean in group High mean in group Low
48.62339          39.22047

> t.test(culmen_length ~ group_culmen_length, data = data, var.equal = TRUE)

Two Sample t-test

data: culmen_length by group_culmen_length
t = 31.413, df = 340, p-value < 2.2e-16
alternative hypothesis: true difference in means between group High and group Low is not equal to 0
95 percent confidence interval:
 8.814141 9.991707
sample estimates:
mean in group High mean in group Low
48.62339          39.22047

> t.test(culmen_length ~ group_culmen_length, data = data, var.equal = FALSE, alternative = "greater")

Welch Two Sample t-test

data: culmen_length by group_culmen_length
t = 31.413, df = 339.97, p-value < 2.2e-16
alternative hypothesis: true difference in means between group High and group Low is greater than 0
95 percent confidence interval:
 8.909215      Inf
sample estimates:
mean in group High mean in group Low
48.62339          39.22047
```

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```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help

Source
Console Terminal Background Jobs
R 4.5.2 ~ /

> t.test(culmen_length ~ group_culmen_length, data = data, alternative = "less")

Welch Two Sample t-test

data: culmen_length by group_culmen_length
t = 31.413, df = 339.97, p-value = 1
alternative hypothesis: true difference in means between group High and group Low is less than 0
95 percent confidence interval:
 -Inf 9.896633
sample estimates:
mean in group High mean in group Low
      48.62339      39.22047

> t.test(culmen_depth ~ group_culmen_depth, data = data)

Welch Two Sample t-test

data: culmen_depth by group_culmen_depth
t = 28.723, df = 320.51, p-value < 2.2e-16
alternative hypothesis: true difference in means between group High and group Low is not equal to 0
95 percent confidence interval:
 3.097797 3.553367
sample estimates:
mean in group High mean in group Low
      18.76534      15.43976

> t.test(culmen_depth ~ group_culmen_depth, data = data, var.equal = TRUE)

Two Sample t-test

data: culmen_depth by group_culmen_depth
t = 28.883, df = 340, p-value < 2.2e-16
alternative hypothesis: true difference in means between group High and group Low is not equal to 0
95 percent confidence interval:
 3.099106 3.552058
sample estimates:
mean in group High mean in group Low
      18.76534      15.43976
```

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The image displays two screenshots of the RStudio interface, showing the results of t-tests performed on a dataset named 'data'.

Top Screenshot:

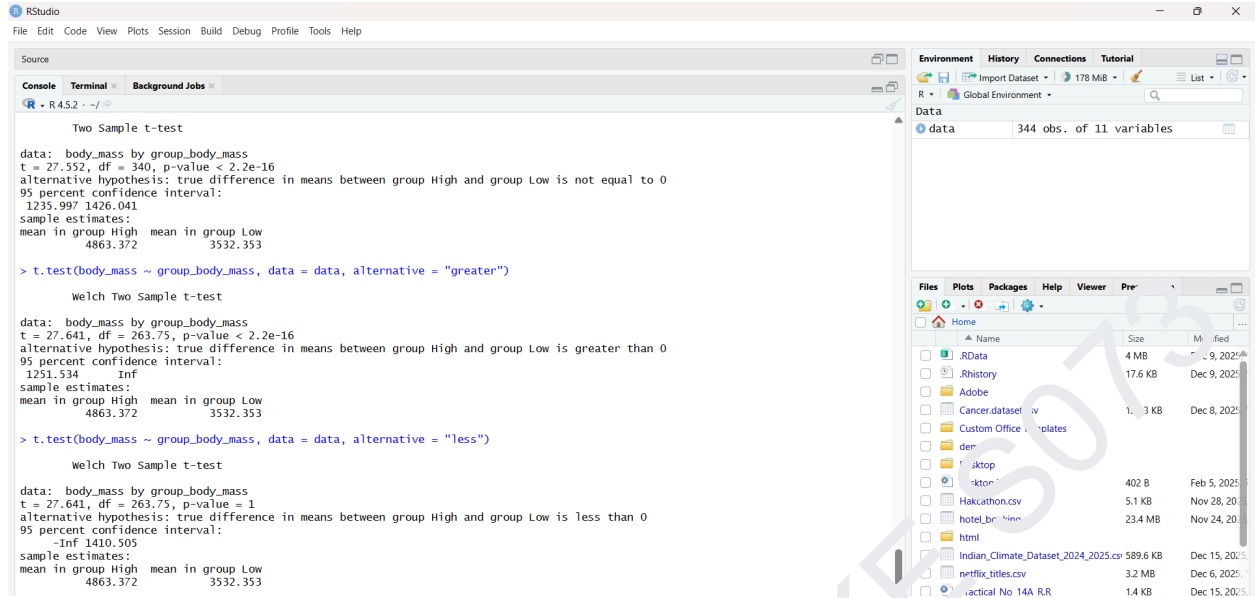
- Console:** Shows the results of two t-tests. The first test is for 'culmen_depth' with the alternative hypothesis 'greater', resulting in a p-value < 2.2e-16. The second test is for 'culmen_depth' with the alternative hypothesis 'less', resulting in a p-value = 1. The third test is for 'flipper_length' with the alternative hypothesis 'not equal to', resulting in a p-value < 2.2e-16.
- Environment:** Shows the 'data' object with 344 observations and 11 variables.

Bottom Screenshot:

- Console:** Shows the results of two more t-tests. The first test is for 'flipper_length' with the alternative hypothesis 'greater', resulting in a p-value < 2.2e-16. The second test is for 'flipper_length' with the alternative hypothesis 'less', resulting in a p-value = 1. The third test is for 'body_mass' with the alternative hypothesis 'not equal to', resulting in a p-value < 2.2e-16.
- Environment:** Shows the 'data' object with 344 observations and 11 variables.

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The screenshot displays the RStudio environment. The console shows the results of three t-tests performed on the 'body_mass' variable, grouped by 'group_body_mass' (High and Low).

Two Sample t-test

```
data: body_mass by group_body_mass
t = 27.552, df = 340, p-value < 2.2e-16
alternative hypothesis: true difference in means between group High and group Low is not equal to 0
95 percent confidence interval:
 1235.997 1426.041
sample estimates:
mean in group High mean in group Low
      4863.372      3532.353
```

> t.test(body_mass ~ group_body_mass, data = data, alternative = "greater")

Welch Two Sample t-test

```
data: body_mass by group_body_mass
t = 27.641, df = 263.75, p-value < 2.2e-16
alternative hypothesis: true difference in means between group High and group Low is greater than 0
95 percent confidence interval:
 1251.534      Inf
sample estimates:
mean in group High mean in group Low
      4863.372      3532.353
```

> t.test(body_mass ~ group_body_mass, data = data, alternative = "less")

Welch Two Sample t-test

```
data: body_mass by group_body_mass
t = 27.641, df = 263.75, p-value = 1
alternative hypothesis: true difference in means between group High and group Low is less than 0
95 percent confidence interval:
      -Inf 1410.305
sample estimates:
mean in group High mean in group Low
      4863.372      3532.353
```

The Environment pane on the right shows a data object named 'data' with 344 observations and 11 variables.

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